

Canonical Mellin Quotient Frames: Finite-Sampling Kernels, Interior Stability, and the Boundary of Signed TPC Reassembly

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Abstract

The preceding twin-prime-correlation reduction left a signed Mellin reassembly between fixed-complexity cell survival and the full projective profile. We determine exactly which frame statement is available. If a continuous Mellin dictionary is allowed arbitrary signed coefficient densities, no lower frame can hold: every synthesis into a finite-dimensional physical Hilbert space has a nonzero smooth kernel. For the phase dictionary at logarithmic samples $y_1, \dots, y_N$, the kernel is explicit,

$$g(t) = \prod_{n=1}^N \left(\frac{d}{dt} + iy_n \right) \rho(t), \quad \rho \in C_c^\infty(\mathbb{R}),$$

and neighboring continuous atoms have arbitrarily large total-variation-to-output condition number. Thus a raw continuous Riesz bound is not merely unproved; it is false.

One provenance-fixed scalar Mellin direction is different at the exact algebraic level. For a fixed smooth shape W , positive samples x_n , and coefficients c_n , Mellin inversion gives the exact identities

$$\mathcal{E}_B(W; c) = A_B(W) \|c\|_2, \quad \|S_W c\|_2^2 = \sum_n |c_n W(x_n)|^2,$$

where $A_B(W)$ is a polynomially weighted Wiener–Mellin mass. Consequently, on a declared sample set on which $\inf_n |W(x_n)| \geq \eta > 0$, the canonical condition number is $O_{B,W,\eta}(1)$. Coordinate samples approaching a zero of the physical cutoff give quantified polynomial, or even super-polynomial, obstructions. For a fixed finite scalar family we bound the quotient condition number in terms of its normalized sampled Gram matrix. Weak convergence of the sample measures and linear independence of the limiting shapes then give an $O(1)$ quotient frame for that abstract fixed-family problem.

We apply this criterion to the prime-squarefree dyadic row universe. Its normalized logarithmic sample measure converges to the measure with density e^{x+y} on $[0, \log 2]^2$. Uniformly for a log-Fourier bank of degree at most a fixed power of $\log X$, the sampled Gram is bounded above and below. The literal TPC static mask changes its lower frame bound by at most

$$O(\epsilon_X (2K + 1)^2), \quad \epsilon_X \ll L^{-1+o(1)} + D^{-1+o(1)} + X^{-\kappa_0+o(1)}.$$

Hence every declared fixed finite scalar family, and every retained finite log-Fourier bank in the stated range, has zero polynomial sampling-and-mask cost at the inherited numerical endpoint. Passing from this theorem to the actual TPC provenance requires an additional bridge: the genuine orbit-translate family is X -dependent and Hilbert-valued, and its weighted continuous Mellin envelope must be compared with the sampled Hilbert energy. Those uniform Gram or relative-tail and norm-comparison hypotheses are formulated here

but are not verified. This bridge therefore remains open before coherent prime summation, residual grouping, physical-shift localization, and native-Gram identification. No fixed-shift prime-pair estimate, parity breach, or twin-prime conclusion is claimed.

Keywords: Mellin inversion; quotient frame; finite sampling; condition number; prime-squarefree rows; static mask; projective reassembly; sieve parity.

Convention. All Hilbert spaces are complex. Continuous synthesis is understood as a Bochner integral. A “canonical” measure always means the fixed real-axis Mellin measure supplied by the physical smooth shapes, before any projective decomposition of the static mask. Constants may depend on the fixed shapes, residue moduli, and derivative order, but not on X . We write $e^z := \exp z$, $e(z) := \exp(2\pi iz)$, and $e_q(z) := e(z/q)$.

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1 Introduction

The TPC reduction has reached a narrow but consequential question. A fixed-complexity residue/Mellin cell survives the literal static row mask with zero polynomial loss [6], whereas the completed TPC-48 estimate is still expressed as a signed outer integral of layerwise Hilbert norms [4]. Can the latter integral be replaced by the energy of the physically reassembled coefficient? Calling this a “frame problem” is useful only after the domain of the putative frame has been specified. The continuous Mellin coefficient space, the finite family of physical cutoff shapes, and the one provenance-fixed coefficient density are three different domains.

This paper separates them. The strongest naive statement is false for an elementary dimensional reason: at every finite scale the physical coefficient space is finite dimensional, while arbitrary smooth signed Mellin densities form an infinite-dimensional space. We give both an abstract smooth kernel and an explicit differential kernel for the phase dictionary. Thus no amount of numerical conditioning evidence can establish a lower Riesz bound on the full continuous parameter space.

The negative theorem does not apply to one fixed Mellin density. Mellin inversion recombines the unique transform density of a fixed cutoff into the values of that cutoff at declared samples. This gives an exact formula, not a heuristic orthogonality assertion. Separately, for a fixed finite scalar family the remaining question is finite dimensional: after exact redundancies are removed, its condition number is controlled by the least eigenvalue of a normalized sampled Gram matrix. We prove that this eigenvalue is bounded away from zero for every fixed limiting-independent scalar family on the prime-squarefree dyadic row universe. We also prove a uniform version for a retained log-Fourier bank of polylogarithmic degree and show that the literal static mask perturbs its lower bound by only $O(\varepsilon_X(2K+1)^2)$. These statements do not by themselves identify the genuine X -dependent, Hilbert-valued orbit-translate family with a fixed scalar family.

1.1 Main results

The results are organized by the logical strength of their domains.

- (R1) **Continuous-domain no-go.** Every integral synthesis from smooth densities on an interval into a finite-dimensional Hilbert space has a nonzero smooth kernel (theorem 3.1). For logarithmic phase samples the kernel is explicit (proposition 3.2), and neighboring atoms give arbitrarily large signed-measure condition number (proposition 3.3).
- (R2) **Exact canonical direction.** For a fixed smooth shape, its weighted Mellin–Wiener envelope and its physical synthesis norm are given exactly by theorem 4.1. A declared sample set satisfying $\inf_n |W(x_n)| \geq \eta > 0$ has $O_{B,W,\eta}(1)$ condition number. Coordinate

samples approaching a finite-order zero incur the corresponding power loss, and a flat cutoff boundary can incur losses beyond every fixed power ([corollary 4.4](#)).

- (R3) **Finite quotient criterion.** For a fixed finite scalar shape family, the squared quotient condition number lies between the inverse least positive eigenvalue and the number of retained shapes times that inverse eigenvalue for its normalized Gram matrix ([theorem 5.1](#)). Weak convergence of sample measures and limiting linear independence therefore give an $O(1)$ quotient frame ([theorem 5.5](#)).
- (R4) **Arithmetic sampling.** The logarithmic empirical measures of admissible prime-squarefree rows converge unconditionally to

$$d\nu(x, y) = e^{x+y} dx dy, \quad 0 \leq x, y \leq \log 2,$$

and their Fourier moments admit a quantitative fixed-modulus estimate ([theorem 6.1](#)). Consequently every fixed limiting-independent scalar family has a stable sampled Gram, and every log-Fourier bank of degree $K \leq (\log X)^A$ obeys uniform Riesz bounds ([theorem 6.2](#)).

- (R5) **Literal-mask robustness for declared finite families.** If a zero-one mask deletes at most ε_X of the complete row universe in every source row, the lower bound of a two-dimensional degree- K bank changes by at most $C\varepsilon_X(2K+1)^2$, uniformly in an arbitrary nonnegative weight on the other side ([theorem 7.1](#)). At the inherited endpoint this is $o(1)$ for every fixed polylogarithmic degree. The fixed finite scalar-family version has constants depending on that family.
- (R6) **Exact boundary and conditional bridge.** The positive result proves the abstract fixed finite scalar-family and retained finite-bank sampled frames. Transfer to the actual coefficientwise pre-terminal provenance requires a uniform normalized Gram or uniform relative Fourier-tail theorem for the X -dependent Hilbert-valued orbit-translate family, together with a norm comparison from the actual weighted continuous envelope to that Hilbert energy. [Proposition 7.4](#) records this criterion and the fact that it is not verified here. The result also does not commute an outer L^1 integral through the nonlinear TPC-48 layer norm and does not control the coherent prime square, residual grouping, physical-shift localization, or native affine Gram.

1.2 Three levels of assertion

To prevent a local Hilbert-space lemma from being promoted into a number-theoretic conclusion, we use the following ledger throughout.

Level L0: exact algebra/analysis.

Kernels, Mellin inversion identities, Gram formulas, and finite-matrix perturbation bounds. These statements do not use primes.

Level L1: arithmetic interface.

Fixed-modulus prime number theory, squarefree progression counts, and the inherited static-mask degree defect. These prove the sampled quotient bounds on the stated row universe.

Level L2: target correlation.

A fixed-shift prime-pair or twin-prime estimate. No result at this level is proved or inferred here.

In particular, no actual TPC provenance gate is called closed merely from the abstract fixed-family theorem. Any future use of that word must first verify the bridge criterion in [proposition 7.4](#); it can never refer to the parity problem or the full TPC determinant route here.

1.3 Organization

[Section 2](#) reconstructs the canonical operator before the projective mask decomposition. [Section 3](#) proves the continuous-domain obstructions. [Section 4](#) treats the single physical direction, and [section 5](#) gives the finite quotient criterion. [Section 6](#) proves the prime-squarefree sampling theorem. [Section 7](#) transfers it through the literal mask. The endpoint and open doors are given in [section 8](#); the reproducible finite certificate is described in [section 9](#).

2 The provenance-canonical synthesis interface

This section reconstructs the exact coefficient interface inherited from TPC-25, TPC-36, and TPC-47-50 [1-6]. There are three different parameter spaces in that interface: a fixed finite residue decomposition, a continuous real-axis Mellin decomposition, and a later projective decomposition of the static mask. Only the first two belong to the provenance-canonical pre-mask cellization. Keeping them separate is essential: a terminal projective parameter from TPC-47/48 is not a physical Mellin atom, and a lower-frame assertion for all of its possible signed representations is false.

2.1 Physical rows and the literal pre-mask coefficient

Fix the physical shift $h_0 \neq 0$ and one source-compatible dyadic packet. Let $\mathcal{A}_1$ and $\mathcal{A}_2$ be its two finite row systems. A row is

$$\rho = (\ell_\rho, d_\rho), \quad m_\rho = \ell_\rho d_\rho \asymp Q, \quad (1)$$

where ℓ_ρ is prime, d_ρ is squarefree, and the inherited separation makes m_ρ determine ρ . The literal static mask is

$$K_{\alpha\gamma} := \mathbf{1}_{\ell_\alpha \neq \ell_\gamma} \mathbf{1}_{|m_\alpha - m_\gamma| > \Delta} \mathbf{1}_{(d_\alpha, d_\gamma) \leq G}, \quad \Delta = QX^{-\kappa_0}, \quad G = X^{\kappa_0}. \quad (2)$$

Before K is decomposed, the joint multiplier has the form

$$\mathfrak{A}_{\alpha,\gamma}^{\text{act}}(j) = K_{\alpha\gamma} \Xi_{\alpha,\gamma}(j) \mathcal{W}_{\alpha,\gamma}(j/J), \quad (3)$$

where Ξ has a fixed orbit period and $\mathcal{W}$ is assembled from fixed smooth row-orbit cutoffs. On the left, the physical source coefficient is

$$\gamma_\alpha^{(1)} = \mu(d_\alpha)(\log \ell_\alpha) \omega_D^{(1)}(d_\alpha) \psi_L^{(1)}(\ell_\alpha/L) \zeta_\alpha^{(1)}, \quad (4)$$

with the symmetric formula on the right. These are literal coefficients, not arbitrary bounded row weights.

The large-prime residual vector at h_0 is

$$\mathcal{F}_{h_0}^{\text{act,L}} = \sum_{\substack{\gamma, j, m, p, r \\ pr - mj = h_0}} \gamma_m^{(1)} \mathfrak{A}_{m,\gamma}^{\text{act}}(j) \mu(r) \Lambda(pr) e_{\gamma, d(m)r} \otimes e_j. \quad (5)$$

It has actual coefficients at the physical intercept, but it belongs to the low-sign-averaged residual Gram route rather than automatically to the native affine physical Gram face. Prime summation, residual grouping, and this Gram identification will therefore remain downstream of the synthesis operator defined below.

2.2 Finite residue variables and continuous Mellin variables

Let $H_0 = O_{h_0}(1)$ be a common orbit-residue period. A fixed residue factor has the finite Fourier expansion

$$\xi(d, j) = \sum_{a, b \bmod H_0} \widehat{\xi}_{H_0}(a, b) e_{H_0}(ad + bj), \quad \sum_{a, b \bmod H_0} |\widehat{\xi}_{H_0}(a, b)| \ll_{h_0} 1. \quad (6)$$

Equivalently one may first split the $O_{h_0}(1)$ physical residue cells. Thus residue parameters are finite and of fixed complexity. Direct cell splitting uses a finite counting measure; Fourier implementation uses a finite complex atomic measure. Neither is a growing continuous frame.

The smooth separation is genuinely continuous. For a fixed smooth shape W put

$$\widetilde{W}(t) := \int_0^\infty W(z) z^{-it} \frac{dz}{z}. \quad (7)$$

The function $\widetilde{W}$ is Schwartz. For the four smooth factors in a two-row packet, a representative provenance measure is

$$\begin{aligned} \boldsymbol{\tau} &= (t_1, s_1, t_2, s_2) \in \mathbb{R}^4, \\ d\nu_{\text{Mel}}(\boldsymbol{\tau}) &:= (2\pi)^{-4} \widetilde{W}_1(t_1) \widetilde{\Psi}_1(s_1) \widetilde{W}_2(t_2) \widetilde{\Psi}_2(s_2) d\boldsymbol{\tau}. \end{aligned} \quad (8)$$

The evident product with further fixed smooth shapes gives the versions used in later packets. In particular, this is a specific complex Borel measure, absolutely continuous with respect to Lebesgue measure, rather than a freely chosen coefficient density. Its polynomially weighted total variation is finite at every fixed order. The associated row phases are unitary: for example,

$$(\ell d)^{it} d^{is} = \ell^{it} d^{i(t+s)}. \quad (9)$$

Combining the fixed residue variables with the required real Mellin variables gives finite disjoint unions

$$\Theta_X^{\text{can}} := \bigsqcup_{a \bmod H_0} \Theta_a, \quad \Theta_a = F_a \times \mathbb{R}^{k_a}, \quad (10)$$

where F_a is fixed and finite and $k_a = O(1)$. The corresponding provenance measures are denoted by λ_a . They are finite complex atomic-continuous product measures obtained using the original Fourier and Mellin normalizations. With no null pairs or algebraic refactorizations inserted, the canonical pre-mask identity is

$$\begin{aligned} &\gamma_\alpha^{(1)} \Xi_{\alpha, \gamma}(j) \mathcal{W}_{\alpha, \gamma}(j/J) \\ &= \sum_{a \bmod H_0} \mathbf{1}_{j \equiv a \pmod{H_0}} \int_{\Theta_a} u_{a, \tau}^L(\alpha) v_{a, \tau}(\gamma) \phi_{a, \tau}(j/J) d\lambda_a(\tau). \end{aligned} \quad (11)$$

There is one compact interval $I_0 \subset \mathbb{R}$, independent of X, a, τ , such that $\phi_{a, \tau} \in C_c^\infty(I_0)$. For every fixed B ,

$$\sum_{a \bmod H_0} \int_{\Theta_a} \|u_{a, \tau}^L\|_\infty \|v_{a, \tau}\|_\infty \|\phi_{a, \tau}\|_{C^B} d|\lambda_a|(\tau) \ll_{B, h_0} X^{o(1)}. \quad (12)$$

This is an upper $L^1(d|\lambda|)$ projective-seminorm estimate. It is not a lower frame bound.

2.3 Why mask parameters are not canonical Mellin cells

Only after (11) is fixed does TPC-36 apply the specified mask representation

$$K_{\alpha\gamma} = \int_{\Omega_{36}} p_\eta(\alpha) q_\eta(\gamma) d\nu_{36}(\eta), \quad \int_{\Omega_{36}} \|p_\eta\|_\infty \|q_\eta\|_\infty d|\nu_{36}|(\eta) = X^{o(1)}. \quad (13)$$

The parameter η is a finite or finitely representable projective-mask parameter. It records same-source orthogonality, signed Ferrers separation, and gcd compression. It is not part of Θ_X^{can} and has no representation-independent physical meaning by itself.

There may also be a further continuous Mellin variable when the smooth terminal factor $\Lambda(pr) = (\log pr)\Phi_0(pr/X)$ is separated. Its measure is again fixed by the real-axis Mellin transform of the declared terminal shape and has Schwartz weighted variation. TPC-47 retains this variable, the prime restriction, and $\mu(r)$ in $A_\omega(p, r, m)$ rather than in the opposite-row pre-mask cell. Accordingly it is a canonical *downstream* Mellin parameter, but it is not an element of Θ_X^{can} in (10). If it is retained as an auxiliary coordinate, its measure is tensorized with λ_X only after the coefficientwise operator below has been formed.

The terminal symbol ω in TPC-47/48 combines a canonical pre-mask cell (a, τ) , a mask parameter η , and any terminal Mellin variables. An individual ω -layer therefore contains $u_{a,\tau}^L p_\eta$ and $v_{a,\tau} q_\eta$, while the literal masked tensor is recovered only after the complete signed η -integration. Appending a cancelling mask-layer pair changes layerwise quantities without changing the physical multiplier. Consequently no canonical cell statistic is assigned to a terminal ω .

2.4 The coefficientwise physical Hilbert output

Let $\mathcal{M}_X$ be the finite set of supported source integers and let $\mathcal{I}_X$ contain the supported pairs (m, r) before coherent prime summation. Let $\mathcal{O}_L$ be the opposite-row set, let $\mathcal{S}_X$ contain the residual labels, and put

$$\mathcal{J}_X := \{j \in \mathbb{Z} : j/J \in I_0\}, \quad \mathcal{V}_L := \ell^2(\mathcal{O}_L \times \mathcal{S}_X), \quad \mathcal{H}_X^{\text{coeff}} := \bigoplus_{(m,r) \in \mathcal{I}_X} \ell^2(\mathcal{J}_X; \mathcal{V}_L). \quad (14)$$

The common compact support I_0 makes $\mathcal{J}_X$ finite for every fixed X . The direct sum deliberately retains (m, r) . It therefore isolates the residue/Mellin synthesis from the later coherent prime square and from the grouping map $s = d(m)r$.

For $\theta = (a, \tau) \in \Theta_X^{\text{can}}$ define the literal-mask layer $k_{X,\theta}^L \in \mathcal{H}_X^{\text{coeff}}$ by

$$k_{X,\theta}^L(m, r, j) := \mathbf{1}_{j \equiv a \pmod{H_0}} u_{a,\tau}^L(m) \phi_{a,\tau}(j/J) \times \sum_{\gamma \in \mathcal{O}_L} K_{m\gamma} v_{a,\tau}(\gamma) e_{\gamma, d(m)r}. \quad (15)$$

Here $u_{a,\tau}^L(m)$ means the value at the unique row determined by m . No $p_\eta q_\eta$ mask layer occurs in (15). Multiplying (11) by the literal K gives the exact coefficientwise profile

$$G_{m,r}^L(j) := \sum_{\gamma \in \mathcal{O}_L} \gamma_m^{(1)} \mathfrak{A}_{m,\gamma}^{\text{act}}(j) e_{\gamma, d(m)r}, \quad (16)$$

$$G_X^L = \sum_{a \bmod H_0} \int_{\Theta_a} k_{X,(a,\tau)}^L d\lambda_a(\tau) \quad \text{in } \mathcal{H}_X^{\text{coeff}}. \quad (17)$$

For each X the arithmetic and orbit supports are finite. Strong measurability and (12) make (17) a genuine Bochner integral. The often-used phrase “signed Bochner integral” includes the complex case: write the polar decomposition

$$d\lambda_X(\theta) = \varepsilon_X(\theta) d\mu_X(\theta), \quad \mu_X := |\lambda_X|, \quad |\varepsilon_X| = 1 \quad \mu_X\text{-a.e.}, \quad (18)$$

where $\lambda_X = \bigoplus_a \lambda_a$. Absorb the polar phase into

$$A_{X,\theta}^L := \varepsilon_X(\theta) k_{X,\theta}^L. \quad (19)$$

2.5 Canonical synthesis, the actual direction, and the reduced Gram

On its natural Bochner domain define

$$\mathcal{S}_X^L f := \int_{\Theta_X^{\text{can}}} f(\theta) A_{X,\theta}^L d\mu_X(\theta) \in \mathcal{H}_X^{\text{coeff}}. \quad (20)$$

For example it is defined whenever $\int |f(\theta)| \|A_{X,\theta}^L\| d\mu_X(\theta) < \infty$. After the fixed projective weights are placed in the measure, the literal physical coefficient is the distinguished direction

$$\boxed{G_X^L = \mathcal{S}_X^L \mathbf{1}.} \quad (21)$$

The coefficient function $\mathbf{1}$ is not a freely selected Mellin density; it records the fixed provenance measure itself.

For the provenance measure at each fixed X , this synthesis is a bounded map from $L^2(\mu_X)$. Indeed, the finite residue part is atomic, the physical output support is finite, the real Mellin phases have unit modulus, and all remaining profile dependence grows at most polynomially in the Mellin variables. Since the density placed in $d\mu_X = d|\lambda_X|$ is Schwartz, one has

$$\int_{\Theta_X^{\text{can}}} \|A_{X,\theta}^L\|_{\mathcal{H}_X^{\text{coeff}}}^2 d\mu_X(\theta) < \infty. \quad (22)$$

Cauchy–Schwarz therefore gives boundedness. Its Gram operator is

$$\Gamma_X^L := (\mathcal{S}_X^L)^* \mathcal{S}_X^L, \quad (23)$$

with integral kernel

$$\mathcal{G}_X^L(\theta, \theta') := \langle A_{X,\theta'}^L, A_{X,\theta}^L \rangle_{\mathcal{H}_X^{\text{coeff}}}. \quad (24)$$

In particular,

$$\|\mathcal{S}_X^L\|_{\mathcal{H}_X^{\text{coeff}}}^2 = \iint \mathcal{G}_X^L(\theta, \theta') d\mu_X(\theta) d\mu_X(\theta'). \quad (25)$$

For every fixed X , all index sets in (14) are finite, so the output space is finite-dimensional, whereas $L^2(\mu_X)$ has a non-atomic continuous Mellin component. Hence $\mathcal{S}_X^L$ has an infinite-dimensional kernel. A lower frame inequality on all of $L^2(\mu_X)$ is therefore impossible for dimensional reasons. The correct coefficient space is the quotient

$$\mathfrak{Q}_X^L := L^2(\mu_X) / \ker \mathcal{S}_X^L, \quad \|[f]\|_{\mathfrak{Q}_X^L} := \inf_{g \in \ker \mathcal{S}_X^L} \|f + g\|_{L^2(\mu_X)}, \quad (26)$$

or, equivalently, the reduced subspace

$$\mathcal{K}_X^L := (\ker \mathcal{S}_X^L)^\perp = \overline{\text{ran}(\mathcal{S}_X^L)^*}. \quad (27)$$

The reduced Gram is

$$\Gamma_{X,\text{red}}^L := \Gamma_X^L|_{\mathcal{K}_X^L}. \quad (28)$$

Its smallest positive spectral value is the exact finite- X lower quotient frame constant. A useful asymptotic theorem must control this value on the specified provenance subspace or must control the single actual direction (21); it must not replace the quotient by the false full-space lower bound.

For later comparison with the inherited Minkowski argument, let $w_B(\theta) \geq 1$ be the fixed polynomial/seminorm weight produced by the C^B orbit profile. Define the canonical displayed-representation envelope

$$\mathcal{E}_{1,B}^L(f) := \int_{\Theta_X^{\text{can}}} w_B(\theta) |f(\theta)| \|A_{X,\theta}^L\|_{\mathcal{H}_X^{\text{coeff}}} d\mu_X(\theta). \quad (29)$$

The actual-direction condition number is

$$\kappa_{\text{can,B}}^L(X) := \frac{\mathcal{E}_{1,B}^L(\mathbf{1})^2}{\|\mathcal{S}_X^L \mathbf{1}\|_{\mathcal{H}_X^{\text{coeff}}}^2}, \quad (30)$$

when the denominator is nonzero. Proving $\kappa_{\text{can,B}}^L(X) = X^{o(1)}$, or the corresponding uniform reduced-Gram bound on a declared finite provenance subspace, is a meaningful representation-efficiency statement. The existence of $\ker \mathcal{S}_X^L \neq \{0\}$ by itself is not a physical obstruction; a genuine negative result must place the actual direction, or a required canonical physical family, in a polynomial near-kernel.

2.6 The full TPC-48 envelope is a different nonlinear object

The terminal projective layer in TPC-48 has, after Fourier transformation in the frozen intercept,

$$B_{\omega,m}(\alpha) := \sum_{p,r} A_{\omega}(p,r,m) e(-pr\alpha) v_{\omega,p,r,m}. \quad (31)$$

For fixed m , distinct residual labels separate the r -square, but at fixed (m,r) the p -sum remains coherent. A fixed admissible global representation $\mathcal{R}$ has the envelope

$$\mathfrak{S}_{\mathcal{R}}(\alpha) := \int C_{\text{LS}}(\phi_{\omega}, H_0)^{1/2} \left(\sum_m \|B_{\omega,m}(\alpha)\|^2 \right)^{1/2} d|\nu|(\omega). \quad (32)$$

Its norm order is exactly:

- (i) the outer projective reassembly is $L^1(d|\nu|)$;
- (ii) the source-row aggregation inside each layer is ℓ^2 in m ;
- (iii) each summand uses the residual Hilbert norm; and
- (iv) the prime sum at fixed (m,r) is formed before that norm is taken.

Moreover $C_{\text{LS}}^{1/2}$ is a profile seminorm weight, not a declared Hilbert norm. The pointwise infimum over admissible representations is not integrated; annular arguments retain one fixed jointly measurable global representation. The resulting field is the completed frozen family, and only its $h = h_0$ slice has literal physical provenance.

Thus (32) is not the norm of $\mathcal{S}_X^L \mathbf{1}$, is not a finite-matrix nuclear norm, and is not the native physical atomic diagonal

$$\mathcal{D}_0 = \sum_i \sum_u |b_{i,u}|^2. \quad (33)$$

The operator (20) isolates only the provenance-canonical residue/Mellin no-cancellation question before coherent prime summation. Even a subpower quotient condition number leaves the coherent prime square, the map $s = d(m)r$, frozen-to- h_0 localization, and the comparison with the native affine Gram face open.

3 Finite-sampling kernels for continuous Mellin synthesis

We first separate a universal continuous-frame question from the provenance-fixed Mellin direction used later. The former has a sharp negative answer for a reason that is independent of arithmetic: at each finite value of X , the physical output space is finite dimensional, whereas a freely weighted continuous dictionary has infinitely many coefficient directions.

3.1 A smooth kernel in every finite-dimensional output

Let $\Theta \subset \mathbb{R}^d$ be open, let $\mathcal{H}$ be a complex Hilbert space of finite dimension N , and let

$$Z : \Theta \longrightarrow \mathcal{H}$$

be continuous. For $g \in C_c^\infty(\Theta)$, define

$$\mathcal{S}_Z g := \int_{\Theta} g(\tau) Z(\tau) d\tau, \quad (34)$$

$$\mathcal{E}_Z(g) := \int_{\Theta} |g(\tau)| \|Z(\tau)\|_{\mathcal{H}} d\tau. \quad (35)$$

The first integral is a Bochner integral. Both quantities are finite because g has compact support.

Theorem 3.1 (Universal smooth-kernel obstruction). *Suppose Z is not identically zero. Then there exists a nonzero $g \in C_c^\infty(\Theta)$ such that*

$$\mathcal{S}_Z g = 0, \quad \mathcal{E}_Z(g) > 0. \quad (36)$$

Consequently, there is no finite constant C for which

$$\mathcal{E}_Z(g) \leq C \|\mathcal{S}_Z g\|_{\mathcal{H}} \quad (37)$$

holds for every $g \in C_c^\infty(\Theta)$.

Proof. Choose $\tau_0 \in \Theta$ with $Z(\tau_0) \neq 0$. By continuity, there is an open ball $U \Subset \Theta$ and a number $m > 0$ such that $\|Z(\tau)\|_{\mathcal{H}} \geq m$ on U . Choose $N + 1$ pairwise disjoint balls $U_k \Subset U$ and nonzero functions $\rho_k \in C_c^\infty(U_k)$. Put

$$w_k := \int_{\Theta} \rho_k(\tau) Z(\tau) d\tau \in \mathcal{H}.$$

The $N + 1$ vectors w_k are linearly dependent, so there are coefficients c_k , not all zero, such that $\sum_k c_k w_k = 0$. The function $g = \sum_k c_k \rho_k$ is nonzero because the supports of the ρ_k are disjoint, and $\mathcal{S}_Z g = 0$. Moreover,

$$\mathcal{E}_Z(g) \geq m \sum_k |c_k| \int_{U_k} |\rho_k(\tau)| d\tau > 0.$$

This proves (36); (37) is then impossible. $\square$

The theorem remains true when $N = N(X)$ grows: it is applied separately at each finite X . It does not use an artificial refactorization of a fixed physical tensor. It states that the coefficient space in (34) cannot be the whole of $C_c^\infty(\Theta)$ if a lower frame is desired.

3.2 An explicit kernel for logarithmic phase samples

The preceding dimension argument can be made completely explicit for the scalar Mellin phase dictionary. For $h \in C_c^\infty(\mathbb{R})$, use the positive-sign Fourier convention

$$\mathcal{F}_+ h(y) := \int_{\mathbb{R}} h(t) e^{ity} dt. \quad (38)$$

Integration by parts gives the sign-sensitive identity

$$\mathcal{F}_+((\partial_t + iy_0)h)(y) = i(y_0 - y)\mathcal{F}_+ h(y). \quad (39)$$

Proposition 3.2 (Explicit phase-sampling kernel). *Let $y_1, \dots, y_N \in \mathbb{R}$, let $a_1, \dots, a_N \in \mathbb{C} \setminus \{0\}$, and put*

$$Z_{\mathbf{y}}(t) := (a_n e^{i t y_n})_{n=1}^N \in \mathbb{C}^N. \quad (40)$$

For every nonzero $\rho \in C_c^\infty(\mathbb{R})$, the function

$$g_{\mathbf{y}, \rho}(t) := \prod_{n=1}^N (\partial_t + i y_n) \rho(t) \quad (41)$$

is nonzero and satisfies

$$\int_{\mathbb{R}} g_{\mathbf{y}, \rho}(t) Z_{\mathbf{y}}(t) dt = 0. \quad (42)$$

Proof. Iterating (39) gives

$$\mathcal{F}_+ g_{\mathbf{y}, \rho}(y) = i^N \prod_{n=1}^N (y_n - y) \mathcal{F}_+ \rho(y). \quad (43)$$

The k -th coordinate of (42) is $a_k \mathcal{F}_+ g_{\mathbf{y}, \rho}(y_k) = 0$.

It remains to check that $g_{\mathbf{y}, \rho} \neq 0$. If it vanished, (43) would imply that $\mathcal{F}_+ \rho(y) = 0$ for every real $y \notin \{y_1, \dots, y_N\}$. The Fourier transform of a compactly supported smooth function extends to an entire function. It would therefore vanish identically, forcing $\rho = 0$, a contradiction. $\square$

There is an equivalent physical-shape formulation. If the logarithmic sample points are y_n , choose a nonzero $F_X \in C_c^\infty(\mathbb{R})$ of the form

$$F_X(s) = \chi(s) \prod_{n=1}^N (s - y_n). \quad (44)$$

Its Fourier transform is Schwartz, while Fourier inversion gives $F_X(y_n) = 0$ at every sample. This example depends on the full sample set, and its derivative seminorms need not be uniform in X . It therefore rules out a theorem uniform over arbitrary smooth shapes; it does not contradict a theorem for the fixed provenance shapes.

3.3 Near kernels and phase aliases

For a finite complex Borel measure ν on Θ , extend the notation by

$$\mathcal{S}_Z \nu := \int_{\Theta} Z(\tau) d\nu(\tau), \quad \mathcal{E}_Z(\nu) := \int_{\Theta} \|Z(\tau)\|_{\mathcal{H}} d|\nu|(\tau). \quad (45)$$

Proposition 3.3 (Neighboring-atom condition number). *Let $I \subset \mathbb{R}$ be an interval on which $Z : I \rightarrow \mathcal{H}$ is continuously differentiable, and suppose*

$$\inf_{t \in I} \|Z(t)\|_{\mathcal{H}} \geq m > 0, \quad \sup_{t \in I} \|Z'(t)\|_{\mathcal{H}} \leq L \quad \text{for some } L > 0. \quad (46)$$

If $t, t + h \in I$ and $h \neq 0$, then for $\nu_{t,h} = \delta_t - \delta_{t+h}$,

$$\frac{\mathcal{E}_Z(\nu_{t,h})}{\|\mathcal{S}_Z \nu_{t,h}\|_{\mathcal{H}}} \geq \frac{2m}{L|h|}, \quad (47)$$

with the ratio interpreted as $+\infty$ when the denominator is zero.

Proof. The two atoms are distinct, so $\mathcal{E}_Z(\nu_{t,h}) = \|Z(t)\| + \|Z(t+h)\| \geq 2m$. The fundamental theorem of calculus gives

$$\|\mathcal{S}_Z \nu_{t,h}\| = \|Z(t) - Z(t+h)\| \leq L|h|.$$

$\square$

Thus continuity of a dictionary is an obstruction, rather than a source of a uniform signed lower frame, when its coefficient measure is free. Narrow smooth bumps centered at t and $t + h$ give the same conclusion up to an arbitrarily small relative error.

For completeness, finite phase sampling also has long-range aliases.

Proposition 3.4 (Simultaneous phase alias). *For the dictionary (40) and every integer $Q \geq 2$, there is an integer q with $1 \leq q \leq Q^N$ such that, for every $t \in \mathbb{R}$,*

$$\|Z_{\mathbf{y}}(t + q) - Z_{\mathbf{y}}(t)\|_2 \leq \frac{2\pi}{Q} \|\mathbf{a}\|_2. \quad (48)$$

Consequently the signed pair $\delta_t - \delta_{t+q}$ has condition number at least Q/π .

Proof. Apply simultaneous Dirichlet approximation to $y_n/(2\pi)$: there is $1 \leq q \leq Q^N$ such that $\|qy_n/(2\pi)\|_{\mathbb{R}/\mathbb{Z}} \leq Q^{-1}$ for every n . Hence

$$|e^{iqy_n} - 1| \leq 2\pi/Q.$$

After factoring out the unit phases e^{ity_n} , summing the squared coordinate bounds proves (48). The envelope of the signed pair is $2\|\mathbf{a}\|_2$. $\square$

The alias q may be extremely large, where a fixed Schwartz Mellin density is tiny. Accordingly, [propositions 3.3](#) and [3.4](#) disprove coefficient-uniform frame statements, but neither proposition estimates the one canonical coefficient measure. That fixed direction is treated next.

4 The fixed canonical Mellin direction

The kernel theorems in [section 3](#) allow the coefficient density to vary. The provenance-canonical representation does not: a fixed physical shape has one fixed real-axis Mellin transform. On orthogonal atomic coordinates its envelope-to-output ratio has an exact elementary formula.

4.1 Wiener–Mellin mass and exact synthesis

Let $W \in C_c^\infty((0, \infty))$. We use the normalization

$$\widetilde{W}(t) := \int_0^\infty W(x) x^{-it} \frac{dx}{x}, \quad (49)$$

$$W(x) = \frac{1}{2\pi} \int_{\mathbb{R}} \widetilde{W}(t) x^{it} dt. \quad (50)$$

For an integer $B \geq 0$, define the polynomially weighted Wiener–Mellin mass

$$A_B(W) := \frac{1}{2\pi} \int_{\mathbb{R}} (1 + |t|)^B |\widetilde{W}(t)| dt. \quad (51)$$

Schwartz decay makes $A_B(W) < \infty$. If $W \neq 0$, then

$$A_B(W) \geq A_0(W) \geq \|W\|_\infty > 0 \quad (52)$$

by (50).

Fix samples $x_1, \dots, x_N > 0$ and a coefficient vector $c = (c_1, \dots, c_N) \in \mathbb{C}^N$. Put

$$v_t(c) := (c_n x_n^{it})_{n=1}^N. \quad (53)$$

The canonical synthesis and its order- B envelope are

$$\mathcal{S}_W c := \frac{1}{2\pi} \int_{\mathbb{R}} \widetilde{W}(t) v_t(c) dt, \quad (54)$$

$$\mathcal{E}_B(W; c) := \frac{1}{2\pi} \int_{\mathbb{R}} (1 + |t|)^B |\widetilde{W}(t)| \|v_t(c)\|_2 dt. \quad (55)$$

Theorem 4.1 (Exact fixed-shape Mellin ledger). *For every W , every positive sample set, and every $c \in \mathbb{C}^N$,*

$$\mathcal{S}_W c = (c_n W(x_n))_{n=1}^N, \quad (56)$$

$$\mathcal{E}_B(W; c) = A_B(W) \|c\|_2. \quad (57)$$

Consequently, whenever $\mathcal{S}_W c \neq 0$, the canonical condition number is exactly

$$\text{cond}_{B,W}(c) := \frac{\mathcal{E}_B(W; c)}{\|\mathcal{S}_W c\|_2} = \frac{A_B(W) \|c\|_2}{\left(\sum_{n=1}^N |c_n|^2 |W(x_n)|^2\right)^{1/2}}. \quad (58)$$

Proof. Mellin inversion gives (56) coordinate by coordinate. Since $|x_n^{it}| = 1$, $\|v_t(c)\|_2 = \|c\|_2$ for every t . This proves (57) and then (58). $\square$

The same proof works with orthogonal Hilbert atoms in place of the standard coordinate vectors and with positive atomic weights absorbed into c_n . It fails after nonorthogonal grouping unless the grouping map is included in the object being estimated.

4.2 Interior stability and the exact worst direction

For a nonempty set $I \subset \{1, \dots, N\}$, let $\mathbb{C}^I$ denote the coefficient vectors supported on I .

Corollary 4.2 (Exact interior minimax). *If $W(x_n) \neq 0$ for every $n \in I$, then*

$$\sup_{0 \neq c \in \mathbb{C}^I} \text{cond}_{B,W}(c) = \frac{A_B(W)}{\min_{n \in I} |W(x_n)|}. \quad (59)$$

In particular, if

$$|W(x_n)| \geq \eta > 0 \quad (n \in I), \quad (60)$$

then every canonical direction supported on I has condition number at most $A_B(W)/\eta$.

If $W(x_{n_0}) = 0$ for some n_0 , then the coordinate vector e_{n_0} is an exact kernel direction. Moreover, after adjoining any direction with nonzero output, the unrestricted condition number is unbounded.

Proof. Formula (58) is bounded above by the right side of (59); equality is attained by a coordinate vector at which $|W(x_n)|$ is minimal. If $W(x_{n_0}) = 0$, then (56) gives $\mathcal{S}_W e_{n_0} = 0$. Adding an arbitrarily large multiple of this kernel vector to one fixed nonzero-output coefficient leaves a nonzero output of fixed size while the envelope tends to infinity. $\square$

Thus a fixed shape has no continuous-frame pathology on a declared interior set where it is bounded away from zero. The relevant condition is lower physical mass, not a lower bound for every Mellin layer.

4.3 Zeros and flat cutoff boundaries

The opposite conclusion holds near a zero of the fixed physical shape. This produces a canonical near kernel without changing the Mellin coefficient measure.

Proposition 4.3 (Finite-order zero obstruction). *Let $x_0 > 0$, let $k \geq 1$, and suppose*

$$W^{(j)}(x_0) = 0 \quad (0 \leq j < k), \quad \sup_{x \in U} |W^{(k)}(x)| \leq M_k \quad (61)$$

on an interval U containing x_0 , where $M_k > 0$. If $x_n \in U$, the interval between x_0 and x_n lies in U , $0 < |x_n - x_0| \leq \Delta$, and $c = e_n$, then either $W(x_n) = 0$, giving an exact kernel direction, or

$$\text{cond}_{B,W}(e_n) \geq \frac{k! A_B(W)}{M_k \Delta^k}. \quad (62)$$

Proof. Taylor's theorem with integral remainder gives

$$|W(x_n)| \leq \frac{M_k}{k!} |x_n - x_0|^k \leq \frac{M_k}{k!} \Delta^k.$$

If the left side is nonzero, insert this bound in (58) with $c = e_n$. If it vanishes, (56) gives the stated exact kernel. $\square$

Corollary 4.4 (Flat-boundary super-polynomial obstruction). *Suppose W extends smoothly by zero across a boundary point $x_0 > 0$ of its support. If a sequence of samples satisfies*

$$0 < |x_X - x_0| \leq X^{-\sigma} \quad (63)$$

for some fixed $\sigma > 0$, then either $W(x_X) = 0$ along a subsequence, producing exact kernel directions there, or, along every subsequence on which $W(x_X) \neq 0$, for every $A > 0$,

$$\text{cond}_{B,W}(e_{x_X}) \gg_{A,B,W,\sigma} X^A. \quad (64)$$

Here e_{x_X} denotes the one-coordinate vector at that sample.

Proof. Every derivative of W vanishes at x_0 . For a prescribed $A > 0$, choose an integer $k > A/\sigma$. If $\sup_U |W^{(k)}| = 0$ on a suitable interval U , then W vanishes there and the sample is an exact kernel for all sufficiently large X . Otherwise apply [proposition 4.3](#) with $\Delta = X^{-\sigma}$. Then $\Delta^{-k} = X^{\sigma k} \geq X^A$, and the fixed constant $k!A_B(W)/M_k$ is absorbed into the implied constant. $\square$

The corollary is a worst-direction statement. It does not say that a boundary region carries a positive fraction of the physical atomic energy. A bounded-overlap partition and an aggregate lower-energy estimate may discard such directions. Conversely, an upper seminorm bound alone cannot do so.

4.4 Scope of the canonical identity

The exact ledger above concerns a Mellin phase acting diagonally on orthogonal atomic coordinates. It does not commute a signed integral through a layerwise nonlinear norm. In particular, it does not by itself compare a Banach profile of the form

$$\int C(\phi_\tau)^{1/2} \|B_\tau\| d|\nu|(\tau)$$

with the norm of the signed reassembled physical coefficient. Such a comparison additionally requires either a coefficient-specific vector frame or a reordering that performs canonical Mellin inversion before the nonlinear large-sieve step. No full TPC-48 profile closure is asserted here.

5 Finite canonical families and quotient frames

A physical decomposition may contain several fixed shapes. Exact duplicates should be removed algebraically, whereas nearly duplicate shapes must be charged through a condition number. This section gives the finite-dimensional criterion that distinguishes the two cases.

5.1 The normalized sampled Gram

Let K be a compact metric space, let $W_1, \dots, W_R \in C(K)$ be nonzero fixed shapes, and choose positive normalization weights

$$A_1, \dots, A_R > 0, \quad D_A := \text{diag}(A_1, \dots, A_R). \quad (65)$$

For the Mellin application, K may be a compact logarithmic box and A_r is the weighted Wiener–Mellin mass of the corresponding fixed physical shape. The abstract argument below uses only positivity of the A_r .

Fix samples $x_1, \dots, x_N \in K$ and positive weights $w_1, \dots, w_N$. Write

$$M := \sum_{n=1}^N w_n \quad (66)$$

and define the physical sampled synthesis $F : \mathbb{C}^R \rightarrow \mathbb{C}^N$ by

$$(Fc)_n := w_n^{1/2} \sum_{r=1}^R c_r W_r(x_n). \quad (67)$$

The declared canonical family envelope is

$$\mathcal{E}_{\text{fam}}(c) := M^{1/2} \sum_{r=1}^R A_r |c_r| = M^{1/2} \|D_A c\|_1. \quad (68)$$

In the Mellin specialization this is exactly the envelope obtained by separating the fixed shapes, applying Mellin inversion to each one, and then using Minkowski's inequality.

Introduce the normalized matrix and Gram matrix

$$\mathbf{B} := M^{-1/2} F D_A^{-1}, \quad G := \mathbf{B}^* \mathbf{B}. \quad (69)$$

Thus, with $d = D_A c$,

$$\frac{\mathcal{E}_{\text{fam}}(c)}{\|Fc\|_2} = \frac{\|d\|_1}{\|\mathbf{B}d\|_2}. \quad (70)$$

If $\ker \mathbf{B} \neq \{0\}$, the right side depends on the chosen representative of one physical output. The canonical quotient norm is

$$\|[d]\|_{1,q} := \inf_{k \in \ker \mathbf{B}} \|d + k\|_1, \quad (71)$$

and the quotient frame condition number is

$$K_q(\mathbf{B}) := \sup_{\mathbf{B}d \neq 0} \frac{\|[d]\|_{1,q}}{\|\mathbf{B}d\|_2}. \quad (72)$$

This removes exact algebraic redundancy; it does not remove a small positive singular value.

Theorem 5.1 (Exact normalized-Gram criterion). *Assume $\mathbf{B} \neq 0$, and let $\sigma_+(\mathbf{B})$ be its smallest positive singular value. If $\lambda_+(G) = \sigma_+(\mathbf{B})^2$ denotes the smallest positive eigenvalue of G , then*

$$\boxed{\frac{1}{\lambda_+(G)} \leq K_q(\mathbf{B})^2 \leq \frac{R}{\lambda_+(G)}}. \quad (73)$$

If G is positive definite, the quotient is trivial and (73) bounds the condition number of the displayed canonical family itself.

Proof. Let $V = (\ker \mathbf{B})^\perp$. Every output has a unique representative $d_V \in V$, and

$$\|[d]\|_{1,q} \leq \|d_V\|_1 \leq \sqrt{R}\|d_V\|_2 \leq \frac{\sqrt{R}}{\sigma_+(\mathbf{B})}\|\mathbf{B}d\|_2.$$

This proves the upper bound. For the lower bound, take a unit right singular vector $v \in V$ belonging to $\sigma_+(\mathbf{B})$. For every $k \in \ker \mathbf{B}$, orthogonality gives $\|v+k\|_2 \geq 1$, and hence $\|v+k\|_1 \geq 1$. Therefore

$$\|[v]\|_{1,q} \geq 1, \quad \|\mathbf{B}v\|_2 = \sigma_+(\mathbf{B}),$$

which proves the lower bound. $\square$

Corollary 5.2 (Subpower quotient criterion). *For a family depending on X , the two hypotheses*

$$R_X = X^{o(1)}, \quad \lambda_+(G_X)^{-1} = X^{o(1)} \quad (74)$$

imply

$$K_q(\mathbf{B}_X)^2 = X^{o(1)}. \quad (75)$$

Conversely, if $\lambda_+(G_X) \leq X^{-\sigma+o(1)}$ for a fixed $\sigma > 0$, then $K_q(\mathbf{B}_X)^2 \geq X^{\sigma+o(1)}$.

The quotient criterion is stronger than necessary for one specified physical coefficient. For a fixed direction d_X with $\mathbf{B}_X d_X \neq 0$, it suffices to bound the single ratio

$$\frac{\|[d_X]\|_{1,q}^2}{d_X^* G_X d_X}, \quad (76)$$

no estimate of the least eigenvalue on unrelated coefficient directions is then required.

5.2 Duplicate and nearly duplicate shapes

Proposition 5.3 (Exact duplicate obstruction). *If two shapes satisfy $W_1(x_n) = W_2(x_n)$ at every sampled point, then the unquotiented physical synthesis F has the nonzero kernel direction $c = e_1 - e_2$. Thus no lower bound for the displayed envelope (68) can hold on all representatives. After quotienting by this direction, the duplicate contributes no physical dimension.*

This is an intrinsic redundancy of the chosen finite family. It is different from appending a null pair to a representation after the canonical family has been fixed.

Proposition 5.4 (Near-duplicate condition lower bound). *For two shapes and the signed coefficient $c = (1, -1)$,*

$$\frac{\mathcal{E}_{\text{fam}}(c)}{\|Fc\|_2} = \frac{M^{1/2}(A_1 + A_2)}{\left(\sum_{n=1}^N w_n |W_1(x_n) - W_2(x_n)|^2\right)^{1/2}}. \quad (77)$$

where the ratio is interpreted as $+\infty$ if the denominator vanishes. In that case c is an exact sampled-kernel direction, as in proposition 5.3. In particular, if

$$\left(\frac{1}{M} \sum_{n=1}^N w_n |W_1(x_n) - W_2(x_n)|^2\right)^{1/2} \leq \varepsilon, \quad (78)$$

then the unquotiented condition number in this direction is at least $(A_1 + A_2)/\varepsilon$.

Proof. Both assertions follow by evaluating (67) and (68) at $c = (1, -1)$. $\square$

The lower bound can be polynomial even though both individual shapes are perfectly nondegenerate. Here is a Mellin-specific example. Suppose K is contained in a fixed compact interval of $(0, \infty)$, $W_1(x) = W(x)$, and $W_2(x) = W(e^\delta x)$, where W is C^1 on a fixed compact interval containing every x_n and $e^\delta x_n$. For $|\delta| \leq 1$, the mean-value theorem in logarithmic coordinates gives

$$|W(e^\delta x_n) - W(x_n)| \leq C_W |\delta|. \quad (79)$$

If $A_1 = A_2 = A_B(W)$, as happens for the weighted Mellin masses of these two dilates, then

$$\frac{\mathcal{E}_{\text{fam}}(1, -1)}{\|F(1, -1)\|_2} \gg_W |\delta|^{-1}. \quad (80)$$

A separation $|\delta| = X^{-\sigma+o(1)}$ therefore creates a genuine polynomial loss. A fixed dyadic separation gives only a fixed constant and is not, by itself, an asymptotic obstruction.

5.3 A positive limiting quotient theorem

The preceding condition is directly verifiable for a fixed physical family. For each X , let $x_{n,X} \in K$ and $w_{n,X} > 0$, put $M_X = \sum_{n=1}^{N_X} w_{n,X}$, and define the probability measure

$$\mu_X := \frac{1}{M_X} \sum_{n=1}^{N_X} w_{n,X} \delta_{x_{n,X}}. \quad (81)$$

Let F_X , B_X , and G_X be the matrices defined above for this sampled set.

Theorem 5.5 (Positive limiting sampled frame). *Let K be a compact metric space and let R be fixed. Suppose μ_X converges weakly to a probability measure μ on K , and suppose $W_1, \dots, W_R \in C(K)$ are linearly independent in $L^2(\mu)$. Define*

$$(G_\infty)_{rs} := \frac{1}{A_r A_s} \int_K \overline{W_r(x)} W_s(x) d\mu(x). \quad (82)$$

Then G_∞ is positive definite and

$$G_X \longrightarrow G_\infty \quad \text{entrywise and in operator norm.} \quad (83)$$

Consequently, for all sufficiently large X ,

$$\lambda_{\min}(G_X) \geq \frac{1}{2} \lambda_{\min}(G_\infty) > 0, \quad K_q(B_X)^2 \leq \frac{2R}{\lambda_{\min}(G_\infty)}. \quad (84)$$

Proof. For $u \in \mathbb{C}^R$,

$$u^* G_\infty u = \int_K \left| \sum_{r=1}^R \frac{u_r}{A_r} W_r(x) \right|^2 d\mu(x).$$

Linear independence makes this positive for $u \neq 0$, so G_∞ is positive definite. Each product $\overline{W_r} W_s$ is continuous on K ; weak convergence gives entrywise convergence. Since R is fixed, entrywise convergence implies operator-norm convergence. Weyl's eigenvalue inequality then gives the first bound in (84), and [theorem 5.1](#) gives the second. $\square$

If the original family is linearly dependent in $L^2(\mu)$, one must first quotient by its fixed limiting relations, choose a basis of the resulting image, and apply the theorem to that reduced family. Merely replacing $\lambda_{\min}$ by the smallest positive eigenvalue of the unreduced finite- X Gram can be misleading: an exact limiting relation may appear at finite X as a small positive eigenvalue.

Weak convergence alone is also insufficient when the number of shapes grows with X . A polylogarithmic Fourier bank requires a quantitative sampling theorem uniform over that bank. The present theorem provides the fixed-family endpoint and the exact Gram criterion; any growing-bank application must verify the additional uniformity explicitly.

Finally, the quotient frame concerns the canonical smooth-shape reassembly on its declared atomic coordinates. It neither estimates a coherent prime sum inside one residual coordinate nor controls residual grouping, frozen-to-physical localization, or identification with the native affine Gram face. In particular, it is not a closure theorem for the full TPC-48 Banach envelope.

6 Prime-squarefree logarithmic sampling

We now verify the finite-Gram hypothesis on the row universe actually used before terminal prime summation. Only standard fixed-modulus prime number theory and Möbius inversion are needed.

Put $T = \log 2$. Fix positive integers q, H , a nonempty set $\mathcal{A} \subset (\mathbb{Z}/q\mathbb{Z})^2$, and $e \in \{0, 1\}$. We assume that every prime residue occurring in $\mathcal{A}$ is reduced modulo q , and retain only those squarefree residue classes having positive local density subject to $(d, H) = 1$. At least one retained pair is required. Define

$$\mathcal{R}_X(\mathcal{A}) := \{(\ell, d) : L < \ell \leq 2L, \ell \text{ prime}, D < d \leq 2D, \mu^2(d) = 1, (d, H) = 1, (\ell, d) \bmod q \in \mathcal{A}\}, \quad (85)$$

$$w_{\ell, d} := (\log \ell)^{2e}, \quad E_X := \sum_{(\ell, d) \in \mathcal{R}_X(\mathcal{A})} w_{\ell, d}. \quad (86)$$

We suppose $X^c \leq L, D \leq X^C$ for fixed $c, C > 0$. The normalized logarithmic empirical measure is

$$\nu_X := \frac{1}{E_X} \sum_{(\ell, d) \in \mathcal{R}_X(\mathcal{A})} w_{\ell, d} \delta_{(\log(\ell/L), \log(d/D))}. \quad (87)$$

For the quantitative comparison it is convenient not to discard the harmless finite- L logarithmic density. For all sufficiently large L , let

$$Z_{L, e} := \int_0^T e^u (\log L + u)^{2e-1} du \quad (88)$$

and define the probability measure

$$d\nu_{L, e}^{\text{ref}}(x, y) := \frac{e^x (\log L + x)^{2e-1}}{Z_{L, e}} dx e^y dy, \quad (x, y) \in [0, T]^2. \quad (89)$$

The second factor already has mass one because $\int_0^T e^y dy = 1$. Notice that the exponent $2e - 1$ equals -1 when the prime row is unweighted; this is the usual density $dt/\log t$ of primes and is the reason for retaining the finite- L reference measure.

Theorem 6.1 (Arithmetic logarithmic sampling). *For every continuous $F : [0, T]^2 \rightarrow \mathbb{C}$,*

$$\int F d\nu_X \longrightarrow \int_{[0, T]^2} F(x, y) e^{x+y} dx dy. \quad (90)$$

More quantitatively, for every fixed $A > 0$ and every $0 < \delta < 1/2$, uniformly when $|\xi| + |\eta| \leq (\log X)^A$,

$$\begin{aligned} & \left| \int e^{i(\xi x + \eta y)} d\nu_X - \int e^{i(\xi x + \eta y)} d\nu_{L, e}^{\text{ref}} \right| \\ & \leq C_{A, \delta, q, H, \mathcal{A}} (1 + |\xi| + |\eta|)^{C_1} \left(e^{-c_1 \sqrt{\log L}} + D^{-1/2 + \delta} \right), \end{aligned} \quad (91)$$

where $c_1 > 0$ and $C_1 < \infty$ are independent of X, ξ, η .

Proof. Split $\mathcal{A}$ into its finitely many residue pairs. For a reduced prime class $a \bmod q$, the fixed-modulus prime number theorem and partial summation give, uniformly for $|\xi| \leq (\log X)^A$,

$$\begin{aligned} & \sum_{\substack{L < \ell \leq 2L \\ \ell \text{ prime} \\ \ell \equiv a \pmod{q}}} (\log \ell)^{2e} e^{i\xi \log(\ell/L)} \\ &= \frac{L}{\varphi(q)} \int_0^T e^x (\log L + x)^{2e-1} e^{i\xi x} dx + O_{A,q} \left(L (\log L)^{2e-1} (1 + |\xi|)^{C_1} e^{-c_1 \sqrt{\log L}} \right). \end{aligned} \quad (92)$$

The same formula at $\xi = 0$ supplies its normalizing mass. The use of the finite- L reference density in (89) removes any artificial $1/\log L$ error from this comparison. Any fixed extra power of $\log L$ arising in partial summation is absorbed by decreasing c_1 , without changing the stated form of the error.

For a locally admissible squarefree class $b \bmod q$, use $\mu^2(d) = \sum_{r^2|d} \mu(r)$ and inclusion–exclusion for $(d, H) = 1$. For each $r \leq \sqrt{2D}$ and each fixed divisor $s \mid H$, the remaining integer is counted in either no class or one class modulo $\text{lcm}(q, s, r^2)$. Summing the resulting $O(1)$ endpoint errors over r , extending the absolutely convergent main-term sum to infinity, and applying partial summation to $e^{i\eta \log(d/D)}$ give

$$\begin{aligned} & \sum_{\substack{D < d \leq 2D \\ d \equiv b \pmod{q} \\ \mu^2(d)=1, (d,H)=1}} e^{i\eta \log(d/D)} \\ &= \mathfrak{c}_{b,q,H} D \int_0^T e^y e^{i\eta y} dy + O_{A,\delta,q,H} \left(D^{1/2+\delta} (1 + |\eta|)^{C_1} \right), \end{aligned} \quad (93)$$

uniformly for $|\eta| \leq (\log X)^A$, with $\mathfrak{c}_{b,q,H} > 0$ precisely for the retained classes. This is the usual squarefree count in a fixed progression with a smooth unit-modulus test factor; no distribution theorem for growing moduli is being used.

Multiply (92) and (93), sum over the fixed set $\mathcal{A}$, and divide by the same expression at $(\xi, \eta) = (0, 0)$. The residue-dependent main-term constants form one positive finite sum, while the x - and y -profiles are common to every retained residue pair. The normalized main term is therefore exactly $\nu_{L,e}^{\text{ref}}$, and the denominator is bounded away from zero after its natural normalization. This proves (91).

For fixed frequencies the normalized prime reference density converges uniformly to e^x . Hence $\nu_{L,e}^{\text{ref}}$ converges weakly to the probability measure $e^{x+y} dx dy$. The quantitative estimate first proves (90) for finite exponential sums with arbitrary real frequencies. This self-adjoint algebra contains the constants and separates points of the closed square, so the Stone–Weierstrass theorem gives the result for every continuous F . $\square$

6.1 A polylogarithmic log-Fourier bank

For an integer $K \geq 0$, let $\Lambda_K = \{-K, \dots, K\}^2$ and $R_K = |\Lambda_K| = (2K+1)^2$. For coefficients $a = (a_k)_{k \in \Lambda_K}$, put

$$P_a(x, y) := \sum_{k=(k_1, k_2) \in \Lambda_K} a_k \exp\left(\frac{2\pi i}{T} (k_1 x + k_2 y)\right). \quad (94)$$

Theorem 6.2 (Polylogarithmic sampled Riesz bounds). *Fix $A > 0$. If $K \leq (\log X)^A$, then for all sufficiently large X , uniformly in a ,*

$$c_A \sum_{k \in \Lambda_K} |a_k|^2 \leq \int |P_a|^2 d\nu_X \leq C_A \sum_{k \in \Lambda_K} |a_k|^2, \quad (95)$$

where $c_A > 0$ and $C_A < \infty$ are independent of K, X . The same statement holds for coefficients in an arbitrary Hilbert space, with $|P_a|^2$ replaced by its squared Hilbert norm.

Proof. For all sufficiently large L , the density in (89) is bounded above and below by positive absolute constants on $[0, T]^2$. Parseval on the square therefore gives

$$c_0 \sum_k |a_k|^2 \leq \int |P_a|^2 d\nu_{L,e}^{\text{ref}} \leq C_0 \sum_k |a_k|^2. \quad (96)$$

Expanding the quadratic form produces frequencies whose coordinates have absolute value at most $4\pi K/T$. By (91), applied with any fixed exponent larger than A so that the harmless constant $8\pi/T$ is absorbed for large X , every entry of the empirical Gram differs from the reference Gram by at most

$$\rho_X(K) \ll_{A,\delta} (1+K)^{C_1} \left(e^{-c_1 \sqrt{\log L}} + D^{-1/2+\delta} \right). \quad (97)$$

The Schur bound, or simply $(\sum |a_k|)^2 \leq R_K \sum |a_k|^2$, shows that the quadratic-form error is at most

$$R_K \rho_X(K) \sum_k |a_k|^2. \quad (98)$$

Because $R_K \ll (1+K)^2$, $K \leq (\log X)^A$, and $D \geq X^c$, this is $o(1) \sum_k |a_k|^2$. Absorb the error into (96). For Hilbert-valued coefficients the scalar Gram is tensored with the identity, so the same spectral bounds apply. $\square$

Corollary 6.3 (Fixed physical families). *Let $W_1, \dots, W_R \in C([0, T]^2)$ be fixed. Quotient their span by the exact relations in $L^2(e^{x+y} dx dy)$, and choose fixed representatives whose nonzero quotient classes are linearly independent. Then the normalized sampled Gram of this reduced family on $\mathcal{R}_X(\mathcal{A})$ has its least eigenvalue bounded away from zero. In particular, its quotient-frame condition number is $O(1)$.*

Proof. Apply theorem 6.1 to each of the finitely many continuous products $W_r \overline{W_s}$. The resulting sampled Gram converges in operator norm to the positive-definite limiting Gram, so theorem 5.5 applies. $\square$

The corollary is the direct tool for the finite collection of fixed physical cutoff shapes. The growing Fourier bank is a stronger stress test and a convenient finite recellization device; it is not being identified with an arbitrary continuous Mellin coefficient space.

7 Sparse-deletion stability of the sampled quotient frame

We now leave the static mask intact and measure its effect directly on the sampled Gram. This is the order fixed in TPC-50; no Fourier–Ferrers–gcd layer of a projective mask representation is declared to be a physical Mellin cell.

Let $\mathcal{R}_X$ be the complete prime-squarefree row universe and let $M_X(\alpha, \gamma) \in \{0, 1\}$ be the literal mask. We use only the inherited degree defect

$$\sup_{\alpha} \#\{\gamma \in \mathcal{R}_X : M_X(\alpha, \gamma) = 0\} \leq \epsilon_X |\mathcal{R}_X|, \quad (99)$$

and the same bound after interchanging the two row systems. In the TPC application,

$$\epsilon_X \ll_H L^{-1+o(1)} + D^{-1+o(1)} + X^{-\kappa_0+o(1)} \quad (100)$$

by the complete-universe degree theorem inherited through TPC-49/50 [5, 6]. For this TPC application we retain the hypotheses under which that degree theorem was proved: $L > 4D$, and the row product $m = \ell d$ determines the prime-squarefree row. The abstract masked frame below applies whenever (99) holds; the citation does not furnish that hypothesis outside this complete-universe regime.

The elementary Nikolskii bound needed below is

$$\sup_{(x,y) \in [0,T]^2} \|P_a(x,y)\|_{\mathcal{H}}^2 \leq R_K \sum_{k \in \Lambda_K} \|a_k\|_{\mathcal{H}}^2, \quad (101)$$

which follows from Cauchy–Schwarz and remains valid for coefficients in any Hilbert space $\mathcal{H}$.

Theorem 7.1 (One-sided masked log-frame). *Fix the admissible row cell, base weights, and degree $K \leq (\log X)^A$ of [theorem 6.2](#). For each source row α , let*

$$P_\alpha(x,y) = \sum_{k \in \Lambda_K} c_{\alpha,k} \exp\left(\frac{2\pi i}{T}(k_1 x + k_2 y)\right), \quad c_{\alpha,k} \in \mathcal{H}, \quad (102)$$

with only finitely many nonzero source rows. Then

$$\begin{aligned} & \sum_{\alpha} \sum_{\gamma \in \mathcal{R}_X(\mathcal{A})} M_X(\alpha, \gamma) w_\gamma \|P_\alpha(x_\gamma, y_\gamma)\|_{\mathcal{H}}^2 \\ & \geq E_X (c_A - C_{A,\mathcal{A}} \epsilon_X R_K) \sum_{\alpha} \sum_{k \in \Lambda_K} \|c_{\alpha,k}\|_{\mathcal{H}}^2, \end{aligned} \quad (103)$$

where (x_γ, y_γ) are the logarithmic coordinates from [\(87\)](#). The corresponding upper bound is

$$\sum_{\alpha} \sum_{\gamma \in \mathcal{R}_X(\mathcal{A})} M_X(\alpha, \gamma) w_\gamma \|P_\alpha(x_\gamma, y_\gamma)\|_{\mathcal{H}}^2 \leq C_A E_X \sum_{\alpha,k} \|c_{\alpha,k}\|_{\mathcal{H}}^2. \quad (104)$$

In particular the lower bound remains positive whenever $\epsilon_X R_K = o(1)$. The assertion is uniform in arbitrary source-row concentration because the estimate is applied before the sum over α .

Proof. The fixed admissible cell has positive density in the complete row universe and its base weights satisfy

$$E_X \asymp_{q,H,\mathcal{A}} |\mathcal{R}_X| (\log L)^{2e}, \quad \max_{\gamma \in \mathcal{R}_X(\mathcal{A})} w_\gamma \ll (\log L)^{2e}. \quad (105)$$

For a fixed α , the contribution removed from the unmasked quadratic form is therefore at most

$$\begin{aligned} & \#\{\gamma : M_X(\alpha, \gamma) = 0\} \max_{\gamma} w_\gamma \|P_\alpha\|_{\infty}^2 \\ & \ll_{q,H,\mathcal{A}} \epsilon_X E_X R_K \sum_{k \in \Lambda_K} \|c_{\alpha,k}\|_{\mathcal{H}}^2, \end{aligned} \quad (106)$$

where [\(99\)](#), [\(105\)](#), and [\(101\)](#) were used in order. Subtract this from the unmasked lower bound in [\(95\)](#), then sum in α . Deleting nonnegative terms can only improve the unmasked upper bound, which proves [\(104\)](#). $\square$

Corollary 7.2 (Fixed-shape quotient under the literal mask). *Let $W_1, \dots, W_R \in C([0, T]^2)$ be a fixed family whose nonzero quotient classes are linearly independent in $L^2(e^{x+y} dx dy)$. For every source row let $F_\alpha = \sum_{r=1}^R c_{\alpha,r} W_r$, with coefficients in an arbitrary Hilbert space $\mathcal{H}$, and assume either finite source support or $\sum_{\alpha,r} \|c_{\alpha,r}\|_{\mathcal{H}}^2 < \infty$. After discarding exact zero classes and choosing an independent quotient basis, there are constants $c, C > 0$ such that*

$$\begin{aligned} c E_X \sum_{\alpha,r} \|c_{\alpha,r}\|_{\mathcal{H}}^2 & \leq \sum_{\alpha} \sum_{\gamma \in \mathcal{R}_X(\mathcal{A})} M_X(\alpha, \gamma) w_\gamma \|F_\alpha(x_\gamma, y_\gamma)\|_{\mathcal{H}}^2 \\ & \leq C E_X \sum_{\alpha,r} \|c_{\alpha,r}\|_{\mathcal{H}}^2 \end{aligned} \quad (107)$$

for all sufficiently large X .

Proof. By [corollary 6.3](#), the unmasked Gram has a uniform positive least eigenvalue. Because the family is fixed,

$$\sup_{[0,T]^2} \left\| \sum_r c_r W_r \right\|_{\mathcal{H}}^2 \leq C_W \sum_r \|c_r\|_{\mathcal{H}}^2.$$

Repeating [\(106\)](#) gives an $O_W(\epsilon_X)$ perturbation, which is absorbed for large X . $\square$

7.1 Canonical Mellin interpretation and the missing bridge

The real-axis row powers in the provenance formula become, after removing global unit phases,

$$\ell^{it_1} d^{it_2} = L^{it_1} D^{it_2} e^{i(t_1 x_\ell + t_2 y_d)}. \quad (108)$$

For one declared fixed scalar function of (x_ℓ, y_d) , integrating these phases against its Mellin transform reconstructs that function exactly; see [theorem 4.1](#). Such a fixed finite scalar family is governed by [corollary 7.2](#). This statement is not yet an identification of the actual orbit-dependent TPC family with that scalar model.

Indeed, the provenance factors include expressions such as $W(m_\gamma j/X)$ and $\Psi(d_\gamma j/K)$. With $z_j = j/J$, their row-log components have the schematic form

$$W(c_X e^{x+y} z_j), \quad \Psi(c'_X e^y z_j), \quad (109)$$

for bounded scale factors on a fixed packet. Retaining the orbit label therefore produces an X -dependent Hilbert-valued row function, not automatically a fixed list of scalar functions on $[0, T]^2$.

Proposition 7.3 (Distinct log translates are independent). *Let $0 \neq w \in C_c^\infty(\mathbb{R})$, and let $a_1, \dots, a_M \in \mathbb{R}$ be distinct. Then the functions $w(\cdot + a_1), \dots, w(\cdot + a_M)$ are linearly independent on $\mathbb{R}$.*

Proof. If $\sum_j c_j w(s + a_j) = 0$, Fourier transformation gives

$$\widehat{w}(\xi) \sum_j c_j e^{ia_j \xi} = 0.$$

The transform $\widehat{w}$ is not identically zero and hence is nonzero on some open interval. The exponential polynomial on the right vanishes on that interval, so it vanishes identically. Taking its first M derivatives at one point gives a Vandermonde system in the distinct a_j , and therefore every $c_j = 0$. $\square$

Applied to $w(s) = W(e^s)$, this proposition shows that smoothness alone cannot collapse the unrestricted orbit translates in (109) to a fixed-rank scalar family. Restriction to the physical log box, support losses, and aggregation in the orbit Hilbert space require a separate uniform argument.

We record exactly what such an argument must supply. For a family $H = (H_\alpha)_\alpha$ of $\mathcal{K}_X$ -valued functions on the log box, write

$$\begin{aligned} \mathcal{Q}_X^0(H) &:= \sum_\alpha \sum_{\gamma \in \mathcal{R}_X(\mathcal{A})} w_\gamma \|H_\alpha(x_\gamma, y_\gamma)\|_{\mathcal{K}_X}^2, \\ \mathcal{Q}_X^M(H) &:= \sum_\alpha \sum_{\gamma \in \mathcal{R}_X(\mathcal{A})} M_X(\alpha, \gamma) w_\gamma \|H_\alpha(x_\gamma, y_\gamma)\|_{\mathcal{K}_X}^2. \end{aligned} \quad (110)$$

Proposition 7.4 (Actual-provenance bridge criterion). *Suppose the coefficientwise actual row field, with source and residual labels retained, has an exact representation*

$$H_{\alpha, X}^{\text{act}}(x, y) = \sum_{r=1}^{R_X} c_{\alpha, r, X} F_{r, X}(x, y), \quad F_{r, X} \in C([0, T]^2; \mathcal{K}_X),$$

after quotienting its exact relations. Let $C_X(c)$ be the declared normalized coefficient energy, including the factor E_X and the fixed shape normalizations. Assume one of the following two uniform alternatives.

(G1) *The actual vector-valued family has a uniform normalized Gram and sparse-deletion estimate:*

$$c_0 \mathcal{C}_X(c) \leq \mathcal{Q}_X^0(H(c)) \leq C_0 \mathcal{C}_X(c), \quad \mathcal{Q}_X^0(H(c)) - \mathcal{Q}_X^M(H(c)) = o(\mathcal{C}_X(c))$$

uniformly on the declared quotient subspace, for fixed $0 < c_0 \leq C_0 < \infty$.

(G2) *There is a retained log-Fourier bank $P_X(c)$ of degree $K_X \leq (\log X)^A$, with bank coefficient energy comparable to $\mathcal{C}_X(c)$, such that the tail is uniformly relative:*

$$\mathcal{Q}_X^0(H(c) - P_X(c)) = o(\mathcal{C}_X(c)).$$

Finally suppose the actual weighted continuous Mellin envelope satisfies the norm comparison

$$(\mathcal{E}_{1,B}^L(\mathbf{1}))^2 \leq X^{o(1)} \mathcal{C}_X(c^{\text{act}}). \quad (111)$$

for a nonzero actual quotient direction with $\mathcal{C}_X(c^{\text{act}}) > 0$. Then, for all sufficiently large X , the masked denominator below is positive and the coefficientwise, pre-terminal actual direction has zero positive polynomial representation loss:

$$\frac{(\mathcal{E}_{1,B}^L(\mathbf{1}))^2}{\mathcal{Q}_X^M(H_X^{\text{act}})} \leq X^{o(1)}. \quad (112)$$

This conclusion retains the source and residual labels and makes no statement about the later coherent prime or grouping maps.

Proof. Under (G1), subtraction gives $\mathcal{Q}_X^M(H(c)) \geq (c_0 - o(1))\mathcal{C}_X(c)$. Under (G2), [theorem 7.1](#) gives the same lower bound for $P_X(c)$, while $\mathcal{Q}_X^M(H - P_X) \leq \mathcal{Q}_X^0(H - P_X) = o(\mathcal{C}_X(c))$. The triangle inequality in the masked L^2 space transfers the lower bound from P_X to H . Combining either lower bound with (111) proves (112). $\square$

Neither alternative nor (111) is verified here for the actual orbit-translate family (109). The fixed scalar corollary and the retained finite-bank theorem prove model cases of the criterion; they do not prove the actual bridge by relabelling.

Most importantly, the source label α remains outside the opposite-row norm in (103). If a later operation coherently sums different source rows before taking a norm, new cross-row Gram terms appear. The theorem neither discards nor estimates those terms.

8 Endpoint ledger, provenance labels, and open doors

We now place the abstract and arithmetic results at the inherited TPC endpoint. This section is a scope ledger: every gain is attached to the operator on which it was actually proved.

8.1 Zero polynomial cost on the abstract sampled quotient face

The endpoint factorization inherited from TPC-50 is

$$Q = X^{267/400+o(1)}, \quad J = X^{133/400+o(1)}, \quad QJ = X^{1+o(1)}, \quad (113)$$

and

$$L = X^{99979/210000+o(1)}, \quad D = X^{10049/52500+o(1)}. \quad (114)$$

In particular the static-mask degree defect satisfies the upper bound

$$\epsilon_X \ll X^{-\eta+o(1)}, \quad \eta := \min \left\{ \kappa_0, \frac{10049}{52500} \right\} > 0. \quad (115)$$

For every fixed $A > 0$ and $K \leq (\log X)^A$,

$$\epsilon_X R_K \ll X^{-\eta+o(1)} = o(1), \quad (116)$$

$$R_K(1+K)^{C_1} D^{-1/2+\delta} \leq X^{-10049/105000+\varepsilon+o(1)} \quad (117)$$

for every fixed $\varepsilon > 0$, after choosing the corresponding sufficiently small $\delta > 0$. Thus the limiting exponent before the arbitrarily small loss is $10049/105000$; choosing $0 < \varepsilon < 10049/105000$ gives a fixed power saving for this squarefree contribution alone. Here ϵ_X is the mask degree defect, $\delta \in (0, 1/2)$ is the auxiliary exponent in the squarefree counting error, and ε is the final arbitrarily small exponent slack. These three quantities are distinct; choosing δ does not alter the definition of η . The prime progression error is smaller than every negative power of $\log X$, even after multiplying by a fixed polylogarithm. Hence the lower bound in (103) remains a fixed positive constant.

Corollary 8.1 (Abstract endpoint sampled-frame stability). *Fix an admissible fixed-modulus prime-squarefree row cell. Let $\mathbf{W} = (W_1, \dots, W_R)$ be a fixed finite scalar family on the log box, let $\mathbf{A} = (A_1, \dots, A_R)$ be fixed positive normalizations, and remove the exact limiting relations. If the resulting limiting normalized Gram has $\lambda_{\min}(G_\infty) =: \lambda_* > 0$, then its sampled Gram and its literal-mask sampled Gram have condition number*

$$O_{q,H,\mathbf{A},\mathbf{W},\mathbf{A},\lambda_*}(1). \quad (118)$$

Thus this fixed scalar-family problem has zero positive polynomial sampling-and-mask loss. Separately, for every fixed $A > 0$, the retained log-Fourier bank with $K \leq (\log X)^A$ has sampled Hilbert-Gram condition number $O_{A,q,H,\mathbf{A}}(1)$, while its displayed ℓ^1 quotient cost is at worst $R_K = X^{o(1)}$. In both statements the source-row label is retained.

Proof. The fixed-family assertion is [theorem 5.5](#) and [corollaries 7.2](#) and [6.3](#); its constants necessarily retain the displayed family, normalization, and limiting-eigenvalue dependence. The growing-bank and literal-mask assertions follow from [theorems 6.2](#) and [7.1](#) together with (116)–(117). $\square$

The phrase “zero polynomial loss” does not mean zero constant loss, does not suppress the dependencies in (118), and does not mean that an actual TPC provenance operator has been identified with this abstract family. It says only that the declared sampled scalar family or retained finite bank consumes no positive exponent.

8.2 What the continuous no-go changes

The universal kernel in [theorem 3.1](#) makes one possible future strategy inadmissible. One may not introduce an arbitrary L^2 or signed L^1 Mellin coefficient density and then claim a lower frame merely because individual phases have unit modulus. The correct alternatives are:

- (a) use the one provenance-fixed density and invert it exactly;
- (b) pass to the quotient by the exact synthesis kernel and specify a quotient norm; or
- (c) retain a separated finite bank and prove its sampled Gram bound.

The present paper proves each alternative only on its declared model domain. It does not identify the actual orbit-translate provenance with the finite scalar family or retained bank; that step is exactly [proposition 7.4](#) and remains unverified.

The boundary obstruction also has two distinct meanings. A uniform coordinatewise diagonal bound fails when an allowed coordinate approaches a zero, and then subdivision or energy selection is needed. An aggregate fixed-family sampled Gram can nevertheless remain stable when a shape has isolated zeros, provided its limiting $L^2(e^{x+y} dx dy)$ quotient classes remain independent. Thus a coordinatewise near kernel is not by itself an aggregate Gram degeneracy, and weighted Mellin mass alone proves neither conclusion.

8.3 Provenance and claim ledger

Object	Status	Exact meaning
Continuous Mellin lower frame	Disproved	False for arbitrary signed smooth densities at every finite X ; the synthesis has a nonzero smooth kernel.
Canonical fixed density	Exact	Mellin inversion reconstructs the physical shape at the sampled rows; no inter-layer orthogonality is assumed.
Fixed finite scalar-shape quotient	Proved on its abstract domain	After exact redundancies are removed, limiting L^2 -independence gives an $O(1)$ sampled Gram condition number, with constants depending on the family and its limiting Gram.
Prime-squarefree log bank	Proved	Fixed modulus, nondegenerate interior block, degree $K \leq (\log X)^A$, with standard PNT and squarefree counts.
Literal static-mask transfer	Proved for the declared families	One-sided row deletion costs $O(\epsilon_X R_K)$; the source row is not coherently collapsed.
Actual orbit-translate provenance bridge	Open	Requires the uniform vector-valued Gram or relative-tail and weighted-envelope comparison in proposition 7.4 ; these are not verified.
Full TPC-48 profile comparison	Open	Its outer $L^1(d \nu)$ occurs after a nonlinear layer norm and includes a coherent prime sum.
Fixed-shift prime pair / twins	Not reached	No parity-breaking estimate or L^2 target correlation is asserted.

8.4 Why the full TPC-48 envelope does not follow

For a fixed representation, the inherited profile has the schematic form

$$\mathfrak{S}_{\mathcal{R}}(\alpha) = \int C_{\text{LS}}(\phi_\omega, H_0)^{1/2} \left(\sum_m \|B_{\omega,m}(\alpha)\|^2 \right)^{1/2} d|\nu|(\omega), \quad (119)$$

where, inside the Hilbert norm, the prime variable has already been summed coherently. The abstract frames proved here act on declared opposite-row log families while retaining the other row and arithmetic labels. Before they can even be placed into the actual provenance operator, [proposition 7.4](#) must be verified. After that bridge, three further noncommutations would still prevent (119) from being replaced by the physical atomic energy:

- (i) the outer total-variation integral is applied after the square root of a layer energy;
- (ii) C_{LS} is a profile seminorm weight, not the norm of a canonical Mellin coefficient; and
- (iii) the coherent prime and residual maps need not intertwine with the opposite-row quotient projection.

Minkowski proves an upper envelope in this order; the quotient theorem does not reverse Minkowski.

8.5 The next mathematical doors

The remaining interfaces, in dependency order, are:

- D1: **Actual-provenance bridge.** For the genuine X -dependent Hilbert-valued orbit-translate family, verify a uniform normalized Gram or a uniform relative Fourier-tail estimate, and prove the weighted continuous-envelope comparison (111). This is not a consequence of the fixed scalar-family theorem.

- D2: **Coherent-prime square.** At fixed source and cofactor, prove a square estimate for $\sum_p A(p, r, m)e(-pr\alpha)v_{p,r,m}$ that is stable under the canonical quotient projection, or give a sharp obstruction.
- D3: **Residual grouping.** Control the map grouping $s = d(m)r$, including its maximal fiber multiplicity and any usable cancellation within a fiber.
- D4: **Physical-shift localization.** Compare the frozen completed family with the single physical intercept h_0 .
- D5: **Native-Gram identification.** Relate the residual low-sign Gram to the native affine physical Gram without discarding cross terms.
- D6: **Parity-sensitive input.** Only after the preceding Hilbert interfaces close would a genuinely new number-theoretic input be able to address a fixed-shift prime-pair lower bound.

Failure at any one door would be informative: it would identify an explicit operator or near-kernel rather than an unspecified appeal to the parity barrier. The abstract positive and translate-obstruction results localize the first remaining door more sharply; they do not assert that this actual-provenance door is already open.

9 Exact finite certificate and claim boundary

The proofs in this paper are analytic. The accompanying certificate is a deterministic regression test for the finite algebra used in those proofs; it is not numerical evidence for the prime number theorem and does not test a twin-prime statistic.

The standard-library Python program

`experiments/tpc51_certificate.py`

uses exact rational arithmetic and performs 18,498 assertions in six groups:

- (C1) 3,825 explicit phase-kernel root identities;
- (C2) 1,260 cofactor null-vector checks for rectangular Vandermonde synthesis matrices;
- (C3) 1,930 boundary-zero and near-duplicate conditioning checks;
- (C4) 1,008 exact positive-definiteness and moment checks for finite sampled monomial Grams;
- (C5) 10,471 principal-minor checks for the sharp dimension-times-deletion perturbation bound on finite Walsh banks; and
- (C6) four exact endpoint fraction identities.

The normalized source hash is

`6c8fac5abc221e00bd885ac59159b08d3a644a39426ff23910a883172d86c78c,`

the semantic payload hash is

`c64511069ce4e590eec2ac973663446bb3a8f6e16f93543daaf5e2334979a332,`

and the certificate-core hash is

`c0be9331cddaec485b3ec560e0368cee89c30f0df3bf42b384c3dd427c09cfe8.`

The committed output is `experiments/tpc51_certificate_output.txt`. Re-running

`python experiments/tpc51_certificate.py -output
experiments/tpc51_certificate_output.txt`

must reproduce it byte for byte when the normalized source hash is unchanged.

9.1 What is certified

The finite checks catch sign errors in the differential phase kernel, failure of the cofactor null-vector identity, accidental singularity of the test Grams, an incorrect sparse-deletion constant, and arithmetic mistakes in the endpoint exponents. Exact rationals avoid tolerance choices and platform-dependent linear algebra.

9.2 What is not certified

The certificate does not prove the analytic uniformity in [theorem 6.1](#); that theorem is proved from the fixed-modulus prime number theorem and the squarefree progression count. It does not enumerate the full physical coefficient system, compute the nonlinear profile in [\(119\)](#), or test cancellation in a coherent prime sum. Most importantly, its finite Gram examples are consistency tests, not a data-driven substitute for the limiting-independence hypothesis of [theorem 5.5](#).

Thus the computational and theorem claims have the same boundary as the rest of the paper: exact L0 algebra and the stated L1 row-sampling interface, with no L2 fixed-shift prime-pair conclusion.

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